G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

II B.Tech II Semester – I MID Examinations APRIL-2024

Academic Year 2023-2024

SUBJECT: Object Oriented Programming through Java
SUBJECT CODE: 124BK
BRANCH: CSE, CSE(AI&ML), CSE(DS)
Date: 16-04-2024(AN)

Max Marks: 30
Time: 2Hrs

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 – Creating

PART-A

(Answer all the questions. Each question carries 1 mark) 1x10=10 Marks

S.No.	Question	CO	BT Level
1			
a)	List any two inbuilt methods of Scanner class.	CO1	L1
b)	Write the syntax of declaring a class and give an example.	CO1	L2
c)	Define data abstraction.	CO1	L1

d)	What will be the output of the following Java program?
	class CharAtExample {
	public static void main(String[] args) {
	String str = "Welcome to GNITS";
	int strLength = str.length();
	System.out.println(str.charAt(0));
	System.out.println(str.charAt(strLength-1));
	}
	}
		CO1	L3

e)	_____ inheritance is the process of deriving a class from another derived class.	CO4	L1
f)	In java, which type of polymorphism is illustrated by method overriding?	CO4	L4
g)	Define package.	CO4	L1
h)	Write the syntax of declaring an abstract class.	CO4	L1

i)	What will be the output of the following Java program?
	class ExceptionHandling
	{
	public static void main(String[] args) {
	try {
	int[] myNumbers = {1, 2, 3};
	System.out.println(myNumbers[10]);
	} catch (Exception e) {
	System.out.println("Something went wrong.");
		CO2	L3

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	} finally {
	System.out.println("The 'try catch' is finished.");
	}
	}
	}		

j)	List any four built-in exception classes in java.	CO2	L1

PART-B

(Answer 04 full questions. Each question carries 5 marks) 4x5=20 Marks

S.No.	Question	Marks	CO	BT Level
2.a)	Discuss about different variations of for loop in java with suitable examples.	[3M]	CO1	L-2
b)	Explain about method overloading in java with an example.	[2M]	CO1	L-2
3.a)	Write a java program that prints the maximum and minimum elements in a one dimensional array.	[3M]	CO1	L-3
b)	Explain about java buzzwords.	[2M]	CO1	L-1
4.a)	Demonstrate the usage of 'final' keyword in java with suitable examples.	[3M]	CO4	L-2
b)	Compare and contrast abstract class and interface in java.	[2M]	CO4	L-4
5.a)	Is it possible to extend an interface? Justify your answer.	[3M]	CO4	L-4
b)	Explain about access modifiers in packages.	[2M]	CO4	L-1
6.a)	Write a java program that describes handling of Arithmetic Exception using try and catch blocks.	[3M]	CO2	L-3
b)	Explain the usage of finally keyword in exception handling using suitable example.	[2M]	CO2	L-2
7.a)	Create an interface named 'Polygon'. The interface consists of an abstract method getArea(). Create two classes 'Rectangle' and 'Triangle' that implement getArea() method to print the area of the respective polygon. Call this method by creating objects of these two classes.	[3M]	CO4	L-3
b)	Explain about inner classes in java using suitable examples.	[2M]	CO1	L-2

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